

Sugar metabolism and metabolite profiling of maize grains are largely improved by the combination of arbuscular mycorrhizal fungi and humic acid

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ABSTRACT

The increased use of chemical fertilizers in agricultural production leads to numerous negative environmental consequences, in addition to having an impact on the quality of the final product. Thus, the increasing demand for sustainable agricultural practices calls for innovative approaches to enhance crop production while minimizing environmental impact. The present study investigated the effects of arbuscular mycorrhizal fungi (AMF) and humic acid (HA) on sugar metabolism and metabolite profiling in maize grains. The study was conducted at the Experimental Farm of Mansoura University, Egypt, over the consecutive 2021 and 2022 growing seasons to test the impact of bio-organic treatments and low doses of chemical fertilizers on maize as a test plant. Combining HA and AMF markedly enhanced the metabolic profile of maize grains, leading to significant increases in total sugars (+16.16 %), total amino acids (+40.94 %), and saturated fatty acids (+145.45 %), while unsaturated fatty acids experienced a slight decrease (-11.11 %). These shifts in primary metabolites, including essential amino acids and fatty acids, are critical for metabolic functions and human nutrition. Additionally, the treatments promoted secondary metabolites accumulation, including water-soluble and lipid-soluble antioxidants, which have an important function in enhancing the antioxidant capacity of maize grains. In addition, the combination of HA and AMF enhanced elemental content in maize grains such as P and K content. Our findings demonstrate that the synergistic use of AMF and HA significantly enhances maize's metabolic profile and nutritive value. This eco-friendly strategy offers a promising path to sustainable maize production, improving food security and environmental health by reducing reliance on synthetic fertilizers. More research on bio-organic fertilizers and plant-soil interactions is needed in various habitats and plants, as well as underlying physiological, biochemical, and molecular studies.

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1. Introduction

Maize (*Zea mays* L.) is one of the world's most significant cereal crops, ranking third behind wheat and rice (Saldívar and Pérez-Carrillo, 2016). In Egypt, maize is considered the second most significant crop, with yearly production reaching roughly 7.1 million tons in 2023 (FAOSTAT, 2024). Maize grains are used for several purposes including fodder, human food, and industrial raw material (Nuss and Tanumihardjo, 2010). Maize is a vital food source for over 20 % of the global population, offering key nutrients like protein, amino acids, starch, unsaturated fatty acids, and antioxidants (Chander et al., 2008; Ranum et al., 2014; Yu et al., 2022). Excessive chemical fertilizer use in maize fields to maximize grain production leads to environmental problems including greenhouse gas emissions, soil acidification, and groundwater pollution, and chemicalized practices are not cost-effective as well (Hao et al., 2019; Guo et al., 2020; Liu et al., 2023). Hence, there is a need to explore eco-friendly strategies for enhancing maize productivity as well as reducing chemical fertilizers usage. Bio-organic fertilizers have lately been utilized to lessen dependency on chemical fertilizers while simultaneously enhancing the growth, productivity, and quality of crops, while also improving their nutritional and processing value (Gao et al., 2020; Ibrahim and El-Sawah, 2022; Sheteiwy et al., 2023). Biofertilizers are compounds that contain diverse types of microorganisms, such as beneficial bacteria and fungi, to promote soil fertility and plant growth (Allouzi et al., 2022). These microorganisms could provide nutrients for plant nutrition through several mechanisms including nitrogen fixation, phosphorus solubilization, potassium releasing, and production of plant growth-promoting substances (Timofeeva et al., 2023). As well as, using microorganisms could enhance the nutritive value and chemical composition of plants (Mahmoud et al., 2024, 2025). Additionally, organic fertilizers are derived from organic sources, such as organic compost, manures, poultry, domestic sewage, and humic acid can be used to fertilize plants. Several studies have shown that utilizing humic acid can improve soil quality, fertilizer utilization rate, and promote crop yield and quality (Ma et al., 2024; Rui et al., 2024; Li et al., 2025; Xu et al., 2025). Therefore, applying bio-organic fertilizers to enhance maize growth, yield, and nutritional value could be a beneficial technique for sustainable agriculture that needs further investigations.

Arbuscular mycorrhizal fungi (AMF) have been previously reported to perform a mutualistic relationship with the roots of maize plants (Gao et al., 2020; El-Sawah et al., 2023). Such a relationship could provide several benefits to maize plant using several mechanisms. AMF could modulate the root architecture by increasing the surface area of the root which in turn could improve nutrient and water uptake. Moreover, improving nutrient transportation especially phosphorus, and nitrogen which can be transferred to host plants by AMF external hyphae (Govindarajulu et al., 2005). In addition, AMF could induce the physiological and biochemical reactions in plant tissues including antioxidant defense system towards environmental stress (El-Sawah et al., 2021; Sheteiwy et al., 2021a, 2021b, 2023; Nader et al., 2024). A recent study by Ma et al. (2022) found an enhancement in root vigor and nutrient accumulation in maize plants, as well as alterations in organic acids in maize root exudates at various growth stages following AMF symbiosis. Moreover, combining AMF with other bio and organic fertilizers increased α -amylase activity, gibberellins, and their transcript levels while decreasing abscisic acid levels in maize grains (Gao et al., 2020). Hence, AMF recently attracted attention to be used as an eco-friendly and inexpensive technique to enhance the growth, productivity, and nutritional value of maize plants.

In addition to AMF, humic acid (HA) is a low-molecular-weight organic compound that helps in improving soil characteristics and plant growth (Ampong et al., 2022). It is mainly formed by animal and plant remains decomposition by soil microorganisms (Liu et al., 2024). HA has been found to improve soil properties such as structure, texture, and water-holding capacity (Li et al., 2019). Moreover, the HA fraction contains around sixty percent of carbon (C), Nitrogen (N), and other

elements such as oxygen (O), hydrogen (H), and sulfur (S), which provide such nutrient sources to soil microorganisms for survival and reproduction (Gao et al., 2020; Sible et al., 2021). Furthermore, plants easily absorb HA, which can enhance vital hormones such as auxin and cytokinin in plants, and it in turn will help enhance nutrients metabolism, photosynthesis, and stress resistance (Ampong et al., 2022). In addition, HA could help in the precipitation of toxic heavy metals which in turn reduce their transportation and intake by plants (Wu et al., 2017). Previous studies recommended HA application for improving maize growth, productivity, and quality (Nunes et al., 2019; Olk et al., 2023; Wyszowski and Kordala, 2024), making HA a promising technique for maize cultivation.

While AMF and HA applications have been studied for their effects on maize growth and productivity (Pinos et al., 2019; Guo et al., 2022; Ouhaddou et al., 2024), there's a crucial void in understanding their impact on grain metabolism at a biochemical level. This study uniquely fills that gap. We're undertaking a deep metabolic profiling to precisely understand how these treatments influence the levels of sugars, organic acids, amino acids, fatty acids, and antioxidants within maize grains. Specifically, we aim to: 1) quantify changes in comprehensive metabolic profiles (primary: sugars, organic acids, amino acids, fatty acids; secondary: antioxidants) in treated maize grains, 2) determine pathways linking primary metabolite accumulation to secondary antioxidant biosynthesis, and 3) correlate these deep metabolic shifts with improved grain quality (total antioxidant capacity and mineral content) under low chemical fertilization. We hypothesize AMF, HA, and their synergy will fundamentally alter maize grain metabolic profiles by optimizing primary-to-secondary metabolite flow. Our rigorous analysis of bioactive compounds will elucidate these biochemical bases, directly contributing to enhanced maize performance.

2. Materials and methods

2.1. Experimental setup

The experiment was conducted at the Experimental Farm of the Faculty of Agriculture, Mansoura University, Egypt, Mansoura (31.0449° N, 31.3537° E). The experimental soil comprised sand (34.17 %), silt (23.53 %), clay (42.30 %), organic matter (2.56 %), nitrogen (147 ppm), phosphorus (16.43 ppm), potassium (316.81 ppm). In this experiment, maize seeds cv. Hageen Fardy No.1 obtained from the Agricultural Research Station in Giza, Egypt were planted on 15 May 2021 and 2022 in a Completely Randomized Block Design (CRBD) with three replications. Each plot area (3 m × 3.5 m) consisted of four ridges (2.5 m) with a 50 cm distance between rows and 25 cm between hills which contain 2–3 grains per hill. The seeds were planted on one side of the ridge and thinning was performed to one plant per each hill after three weeks from planting. During both seasons, irrigation was performed every two weeks, and weeds were hand-pulled if necessary. The plants were then grown in four experimental settings: (1) Control; (2) AMF; (3) HA; and (4) AMF+HA.

2.2. Mycorrhizal inoculation

The AMF spores used were a mixture of different spores of *Glomus clarum*, *Funneliformis mosseae*, and *Gigaspora margarita*. These were collected from the Agricultural Microbiology Department, Faculty of Agriculture, Mansoura University, Egypt. The AMF inoculum was prepared as described by El-Sawah et al. (2021). Briefly, AMF spores were grown for six months on the roots of *Sorghum sudanenses* Pers., then to each plant, 5 g of trapped soil was added (approx. 50 spores per 1 g soil). To produce mycorrhizal treatment, AMF inoculum was inserted at a depth of 5 cm below the soil surface before sowing. The non-mycorrhizal treatment received equal amounts of autoclaved inoculum to guarantee the supply of the same nutrients.

2.3. Humic acid application

Humic acid (85 %) was obtained from Sphinx International Trade Company, Cairo, Egypt (contains 56 % C, 4.5 % H, 31 % O, and 4.5 % N). HA (12 Kg/hectar) was added along with the first dose of chemical fertilizers according to the study of [Gao et al. \(2020\)](#).

2.4. Chemical fertilization

The nitrogen fertilizer was applied at the rate of 285 kg/ha as urea (46 % N) in two equal doses; the first dose was applied after thinning, and the second was applied before the second irrigation. Throughout the soil preparation process, 476 kg/ha of calcium superphosphate (15.5 % P_2O_5) was applied as P fertilizer. Potassium fertilizer was applied at the rate of 120 kg/ha as potassium sulfate (48 % K_2O) after thinning along with the first dose of nitrogen. The bio and organic fertilizer treatments (AMF, HA, and AMF+HA) received 50 % of the recommended NPK doses, while the control plants received the full dose of NPK.

2.5. Elemental analysis

Plant mineral analysis was performed by first digesting dried, ground tissue samples using a microwave-assisted acid digestion protocol (HNO_3 and H_2O_2 , (5:1 v/v)). The resulting digests were diluted with deionized water and filtered before analysis. Inductively Coupled Plasma Mass Spectrometry (ICP-MS) was then used to quantify a range of plant minerals based on the protocol previously described by [Abdelgawad et al. \(2022\)](#). For the detection of soil minerals soil samples were air-dried, sieved, and digested using a microwave-assisted acid digestion protocol (e.g., HNO_3 and HF for total element analysis). For available nutrient fractions, standard extractants such as DTPA and ammonium acetate were used. The resulting solutions were filtered and analyzed using Inductively Coupled Plasma Mass Spectrometry (ICP-MS) to determine concentrations of minerals.

2.6. Determination of sugars in grains

Individuals soluble sugars were extracted in 50 mM TAE buffer (pH 7.5) containing a mixture of polyclar (0.15 %), Na azide (0.02 %), PMSF (2 mM), mercapto-ethanol (1 mM), mannitol (10 mM), and $NaHSO_3$ (12 mM) ([Naudts et al., 2014](#)). The mixture was centrifuged at 15,000 $\times$ g, 4 °C, for 10 min, and the mixture was heated at 90 °C for 5 min, followed by incubation at room temperature ([Vergauwen et al., 2000](#)). Total soluble and total insoluble sugar concentrations were determined via an adapted ([Naudts et al., 2014](#)). Frozen grains (50 mg) was homogenized in 2 mL acetone to remove pigments and soluble sugars, then centrifuged. The residue was subsequently acid-extracted (1.1 % HCl, 100 °C for 30 min) for insoluble sugars. After dilution and anthrone reagent addition, absorbance was measured at 620 nm and quantified using a glucose standard curve. Following soluble sugar extraction, the pellet was treated with α -amylase and amyloglucosidase (100 U/mL, 1 h, 45 °C) to extract starch ([Galtier et al., 1995](#)). Total soluble and starch sugar concentrations were then quantified using the anthrone reagent method. For cellulose determination, 0.2 g of dried plant material was boiled in 100 °C nitric acid/acetic acid (1:8, v/v, 1 h) to remove lignin, hemicellulose, and xylosans. After successive centrifugations (14,000 $\times$ g, 15 min), the residue was diluted with 67 % H_2SO_4 (v/v), and cellulose was measured at 620 nm using the anthrone reagent.

2.7. Determination of organic, amino, and fatty acids in grains

Organic acids, amino acids, and fatty acids were determined using the protocol described by [Sheteiwy et al. \(2023\)](#). Briefly, organic acids were detected in tested plants using HPLC under the following conditions (0.001 N sulfuric acid, at 210 nm, 0.6 mL min⁻¹). The HPLC system consisted of a liquid chromatograph (Dionex, USA) and detector (LED,

ultimate 3000), in addition to a pump (LPG-3400A), a column thermostat (TCC-3000SD) and an autosampler (EWPS-3000SI). Separation of organic acids was conducted through an Aminex HPH-87H (300 $\times$ 7.8 mm) column coupled with an IG Cation H (30 $\times$ 4.6) precolumn (Bio-Rad) at 65 °C. Data analysis and interpretation were performed using Chromeleon software (v.6.8). Extraction of amino acids was performed by grinding 100 mg of plant sample in 5 mL of 80 % ethanol, followed by centrifugation at 4000 $\times$ g for 25 min. Fatty acid extraction was performed using 200 mg of plant sample. Fatty acids were detected in treated and non-treated plants by using GC/MS (Hewlett Packard, USA) ([Alotaibi et al., 2021](#)). Fatty acids were initially extracted from 0.1 g of dried, powdered sprouts by incubation in 50 % aqueous methanol at 25 °C. Subsequent identification was carried out using GC-MS analysis, employing a Hewlett Packard 6890 gas chromatograph linked to an MSD 5975 mass spectrometer and an HP-5 MS column (30 m $\times$ 0.25 mm $\times$ 0.25 mm). Compounds were positively identified through comparison with spectral data from the Golm Metabolome Database (<http://gmd.mpimp-golm.mpg.de>) and the NIST 05 database.

2.8. Determination of antioxidant metabolites in grains

Total antioxidants were extracted from 200 mg of the frozen plant tissue. Trolox (supplier, location; 0–650 μ M) was used as standard, and total non-enzymatic antioxidant activity was quantified as (μ mol Trolox/g FW) according to the method of [Benzie and Strain \(1996\)](#). Total polyphenol content was extracted from 100 mg of the frozen tissue in 80 % ethanol (v/v) and determined using a Folin-Ciocalteu assay according to the micro-plate method of ([Zhang et al., 2006](#)). An updated aluminum chloride colorimetric method was used to determine flavonoid contents, with quercetin used as a standard ([Sheteiwy et al., 2017](#)).

Ascorbate (ASC) and glutathione (GSH) were determined in the frozen plant tissue by the HPLC method by following the protocol of [Potters et al. \(2004\)](#). Total antioxidant concentration was determined after reducing with 0.04 M DTT and the redox status was calculated as the ratio of the reduced form and the total concentration of the antioxidant. Tocopherol content was determined by the HPLC method. 100 mg of frozen tissue was ground using the MagNAlyser and tocopherols were extracted by using hexane ([Potters et al., 2004](#)).

2.9. Statistical analysis and experimental design

Data collected across both growing seasons were combined to provide a robust dataset for analysis. A one-way analysis of variance (ANOVA) was then performed using SPSS v25.0 (SPSS, Inc., Chicago, IL, USA) to identify significant differences among the various treatment groups for all measured parameters. Following significant ANOVA results, Duncan's multiple-range tests were applied at a probability level of $P \leq 0.05$ to separate and compare individual treatment means. This allowed for a detailed understanding of which specific treatments yielded statistically distinct effects. To further explore patterns and relationships within the extensive metabolic profiling data, principal component analysis (PCA) was conducted using RStudio. PCA is a powerful multivariate statistical tool that reduces the dimensionality of complex datasets while retaining most of the variation, making it ideal for visualizing similarities and differences among samples based on their metabolic fingerprints.

3. Results

3.1. The effects of AMF, HA, and their combination on the contents of sugar metabolites

AMF treatment significantly enhanced sucrose and lactose accumulation compared to the control group ([Table 1](#)). Similarly, HA treatment independently increased the concentrations of glucose and fructose. However, the combined application of AMF+HA exhibited synergistic

Table 1
The contents of sugars in maize grains as affected by AMF, HA, and their combination.

Treatments	Sugars in grains (mg /g FW)						
	Glucose	Fructose	Sucrose	Raffinose	Lactose	Fructose	Total Sugars
Control	2.26 ± 0.25b	1.32 ± 0.22b	2.30 ± 0.21b	0.37 ± 0.03c	0.16 ± 0.01b	0.55 ± 0.04b	5.88 ± 0.54b
AMF	2.02 ± 0.19b	1.09 ± 0.25b	5.41 ± 0.94a	0.66 ± 0.12b	0.36 ± 0.06a	1.07 ± 0.19a	5.35 ± 0.64b
HA	3.47 ± 0.47a	2.42 ± 0.37a	2.24 ± 0.20b	0.47 ± 0.06c	0.16 ± 0.02b	0.66 ± 0.08b	11.34 ± 1.58a
AMF+HA	3.74 ± 0.44a	2.77 ± 0.23a	5.46 ± 0.74a	0.83 ± 0.10a	0.37 ± 0.05a	1.27 ± 0.15a	11.93 ± 1.53a
	Cellulose	Hemicellulose	Fructan	Starch	Structural sugars	Total sugars	
Control	191.78 ± 30.79bc	68.36 ± 3.21c	94.93 ± 12.99b	232.53 ± 15.99b	366.67 ± 36.74a	450.46 ± 14.42c	
AMF	156.34 ± 9.95c	75.53 ± 12.71bc	112.5 ± 26.93ab	156.34 ± 9.95c	331.33 ± 108.01a	558.07 ± 3.24a	
HA	268.62 ± 22.27a	87.19 ± 4.80b	112.08 ± 5.78ab	268.62 ± 22.27a	386.33 ± 26.01a	467.83 ± 24.06c	
AMF+HA	216.62 ± 10.52b	100.89 ± 4.0a	142.51 ± 9.10a	216.62 ± 10.52b	417.33 ± 83.99a	523.26 ± 22.43b	

Data included are means ± SD; different letters within the same column indicate significant differences between means after Duncan’s test ($P \leq 0.05$). AMF, Arbuscular mycorrhizal fungi; HA, Humic acid.

effects, resulting in a significantly higher accumulation of these metabolites compared to individual treatments. Sucrose content showed marked elevation following individual treatments, yet the combined AMF+HA application demonstrated superior efficacy (Table 1). This synergistic effect was most pronounced during these stages, though the differences diminished at physiological maturity. The accumulation patterns of glucose and fructose displayed similar responses. While individual treatments enhanced their concentrations, the combined AMF+HA treatment resulted in significantly higher levels.

Raffinose accumulation also exhibited significant enhancement under both individual treatments, with the combined AMF+HA application yielding the most pronounced and significant effect, reaching a maximum value of 0.83 (mg/g FW), compared to 0.66 (mg/g FW) and 0.47 (mg/g FW) under AMF and HA treatments, respectively. Cellulose and hemicellulose remained relatively unaffected by both individual and combined treatments throughout grain development, suggesting a limited influence of these combined applications of AMF+HA on structural sugar metabolism.

Furthermore, fructan and starch metabolism showed differential and highly significant responses to the treatments. While individual applications of AMF and HA enhanced fructan accumulation, the combined treatment demonstrated significantly higher efficacy, reaching a maximum value of 142.51 (mg/g FW) compared to 112.5 (mg/g FW) and 112.08 (mg/g FW) under the individual treatments of AMF and HA, respectively (Table 1). Overall, Total sugar concentration represents the sum of the individually measured total insoluble and total soluble sugar fractions. Importantly, the combined application of AMF and HA treatments led to a notable increase in these total sugar levels, indicating an enhanced accumulation of carbohydrates within the maize grains. This suggests that these bio-organic amendments positively influenced overall sugar metabolism.

3.2. The effects of AMF, HA, and their combination on the contents of fatty acids

Individual AMF treatment significantly altered the content of saturated fatty acids. We observed marked increases in myristic, and tricosanoic, pentacosanoic acid concentrations compared to the control (Table 2). Additionally, HA treatment independently demonstrated an increase in all saturated fatty acids. Remarkably, the combined AMF+HA application resulted in significantly higher accumulation of myristic, palmitic, stearic, and tricosanoic compared to either individual treatment, reaching a maximum value of (0.99, 46.35, 4.03, and 0.10 mg/g FW), compared to (0.83, 28.10, 2.23, and 0.04 mg/g FW) and (0.66, 33.34, 2.51 and 0.08 mg/g FW) under the individual treatments of AMF and HA, respectively.

Regarding unsaturated fatty acids, palmitoleic, heptadecenoic, oleic, linoleic, eicosenoic, erucic and tetracosenoic acids displayed significant responses to the treatments. HA individually enhanced the content of palmitoleic and linolenic acids, while AMF alone enhanced the content of erucic acid. The combined AMF+HA treatment exhibited synergistic effects, significantly increasing the concentration of palmitoleic, linoleic, and eicosenoic acids compared to individual treatments (Table 2). Notably palmitoleic, linoleic, and eicosenoic acid levels were significantly elevated under the combined treatment (0.21, 0.35, and 2.72 mg/g FW), compared with the individual treatments of AMF (0.10, 0.20, and 1.10 mg/g FW) and HA (0.19, 0.28, and 1.66 mg/g FW), respectively demonstrating the most pronounced response to the synergistic effects of AMF+HA.

Overall, the synergistic effect was particularly evident in the enhanced accumulation of fatty acids.

Table 2
The contents of fatty acids in maize grains as affected by AMF, HA and their combination.

Treatments	Fatty acids in grains (mg /g FW)								
	Myristic (C14:0)	Palmitic (C16:0)	Heptadecanoic (C17:0)	Stearic (C18:0)	Arachidic (C20:0)	Docosanoic (C22:0)	Tricosanoic (C23:0)	Pentacosanoic (C25:0)	Saturated
Control	0.54 ± 0.07c	17.97 ± 2.55c	0.03 ± 0.00b	1.63 ± 0.24b	1.25 ± 0.14b	0.56 ± 0.06b	0.03 ± 0.00d	0.002 ± 0.00c	22.00 ± 3.01c
AMF	0.83 ± 0.13ab	28.10 ± 5.89bc	0.06 ± 0.01b	2.23 ± 0.30b	1.34 ± 0.21b	0.65 ± 0.07b	0.04 ± 0.00c	0.004 ± 0.00b	33.25 ± 6.58bc
HA	0.66 ± 0.1bc	33.34 ± 4.03b	0.13 ± 0.02a	2.51 ± 0.38b	1.90 ± 0.26a	1.04 ± 0.14a	0.08 ± 0.01b	0.013 ± 0.00a	39.66 ± 4.89b
AMF+HA	0.99 ± 0.15a	46.35 ± 8.12a	0.14 ± 0.02a	4.03 ± 0.72a	2.10 ± 0.43a	1.20 ± 0.11a	0.10 ± 0.01a	0.013 ± 0.00a	54.92 ± 9.53a
	Palmitoleic (C16:1)	Heptadecenoic (C17:1)	Oleic (C18:1)	Linoleic (C18:2)	Linolenic (C18:3 ω-3)	Eicosenoic (C20:1)	Erucic (C22:1)	Tetracosenoic (C24:1)	Unsaturated
Control	0.08 ± 0.01b	1.98 ± 0.36a	23.74 ± 3.69a	0.15 ± 0.02c	0.02 ± 0.00b	0.97 ± 0.10c	0.06 ± 0.01c	0.009 ± 0.00c	27.00 ± 4.12a
AMF	0.10 ± 0.01b	1.26 ± 0.15b	12.81 ± 1.74b	0.20 ± 0.02c	0.02 ± 0.00b	1.10 ± 0.10c	0.17 ± 0.09a	0.012 ± 0.00c	15.66 ± 2.03b
HA	0.19 ± 0.03a	2.10 ± 0.17a	23.31 ± 3.59a	0.28 ± 0.04b	0.03 ± 0.00a	1.66 ± 0.16b	0.10 ± 0.02ab	0.022 ± 0.00b	27.69 ± 3.90a
AMF+HA	0.21 ± 0.02a	2.10 ± 0.26a	18.97 ± 3.82ab	0.35 ± 0.05a	0.03 ± 0.00a	2.72 ± 0.23a	0.14 ± 0.02ab	0.033 ± 0.00a	24.56 ± 4.25a

Data included are means ± SD; different letters within the same column indicate significant differences between means after Duncan’s test ($P \leq 0.05$). AMF, Arbuscular mycorrhizal fungi; HA, Humic acid.

3.3. The impact of AMF, HA, and their combinations on the contents of amino acids

AMF treatment demonstrated significant impact on enhancing specific amino acids compared to the combined treatment. Notably, lysine and alanine concentrations exhibited the highest increases under individual AMF application (6.46 and 12.82 mg/g FW, respectively), with their levels being significantly higher than those observed in the combined AMF+HA treatment (4.72 and 11.83 mg/g FW, respectively) (Table 3). HA treatment independently showed remarkable effects on several amino acids. Histidine, methionine, threonine, valine, and asparagine concentrations were significantly elevated under HA treatment, exhibiting higher accumulation (2.24, 0.23, 1.04, 1.79, and 3.48 mg/g FW, respectively) compared to both AMF and combined treatments. This suggests a specific response pathway activated by HA for these amino acids.

Conversely, isoleucine content decreased significantly across all treatment groups except for AMF compared to the control, indicating a negative impact of individual HA and combined treatments on its accumulation. leucine concentrations showed no significant variations across any treatment groups, including the combined AMF+HA application, suggesting its metabolism remained largely unaffected by these treatments (Table 3). The differential responses of amino acids to individual and combined treatments indicate specific metabolic pathways being activated or suppressed by AMF and HA.

3.4. The impact of AMF, HA, and their combinations on elemental composition

Strikingly, phosphorus (P) content in maize grains significantly decreased under AMF application compared to the control (Table 4). This reduction was further and significantly amplified by HA treatment. However, the combined AMF+HA treatment led to a significant recovery in P content, increasing levels notably above both individual treatments and the control. This strongly indicates that while individual applications reduced P, their combination profoundly enhanced its accumulation during grain development.

In contrast, potassium (K) content consistently exhibited a significant increasing trend throughout the experimental period. AMF treatment induced a more significant increase in K than HA treatment (Table 4). The maximum K increase occurred with the combined AMF+HA treatment, reaching significantly higher levels than the control and individual applications. This highlights that the combined application particularly and significantly promoted K accumulation in maize grains.

For Ca, Mg, Na, Fe, Mn, and Zn content, there was a differential response of these elements to the treatment groups compared to the control (Table 4). Overall, these findings demonstrate that AMF and HA applications specifically and significantly influenced only the P and K elemental composition of maize grains.

Table 3

The contents of amino acids in maize grains as affected by AMF, HA and their combination.

Treatments	Amino acid in grains (mg /g FW)								
	Glycine	Proline	Lysine	Histidine	Alanine	Arginine	Ornithine	Glutamine	Asparagine
Control	12.00 ± 0.16b	45.87 ± 0.6b	4.08 ± 0.97c	1.05 ± 0.02d	9.04 ± 1.95b	0.85 ± 0.01b	1.70 ± 0.02b	43.21 ± 0.57c	1.51 ± 0.02d
AMF	15.28 ± 1.47a	60.6 ± 4.79a	6.46 ± 0.96a	1.90 ± 0.19b	12.82 ± 0.17a	1.18 ± 0.10a	2.30 ± 0.12a	54.70 ± 0.71b	2.67 ± 0.04b
HA	13.81 ± 0.18ab	64.73 ± 2.68a	5.58 ± 0.08ab	2.24 ± 0.03a	12.34 ± 1.92a	1.12 ± 0.09a	2.61 ± 0.24a	53.23 ± 2.72b	3.48 ± 0.05a
AMF+HA	15.78 ± 2.11a	65.83 ± 1.74a	4.72 ± 0.07bc	1.59 ± 0.21c	11.83 ± 0.15a	1.11 ± 0.02a	2.57 ± 0.44a	58.54 ± 0.76a	2.52 ± 0.04c
	Isoleucine	Leucine	Methionine	Threonine	Valine	Serine	Phenylalanine	Glutamic Acid	
Control	0.60 ± 0.01b	0.16 ± 0.00a	0.15 ± 0.01c	0.34 ± 0.11d	0.57 ± 0.21d	0.46 ± 0.04a	2.32 ± 0.03a	3.54 ± 0.98a	
AMF	0.89 ± 0.01a	0.21 ± 0.10a	0.22 ± 0.00b	0.73 ± 0.04b	1.37 ± 0.02b	0.58 ± 0.21a	3.18 ± 0.05a	4.79 ± 0.97a	
HA	0.56 ± 0.01bc	0.19 ± 0.02a	0.23 ± 0.00a	1.04 ± 0.02a	1.79 ± 0.19a	0.59 ± 0.06a	3.60 ± 1.03a	4.64 ± 0.97a	
AMF+HA	0.53 ± 0.04c	0.21 ± 0.01a	0.22 ± 0.00b	0.59 ± 0.01c	0.98 ± 0.34 cd	0.65 ± 0.25a	3.45 ± 1.03a	4.72 ± 0.06a	

Data included are means ± SD; different letters within the same column indicate significant differences between means after Duncan's test ($P \leq 0.05$). AMF, Arbuscular mycorrhizal fungi; HA, Humic acid.

3.5. The effects of AMF, HA, and their combination on antioxidants

TAC levels significantly increased under HA treatment compared to AMF, and the highest TAC value was observed under the combined AMF+HA treatment 230.10 $\mu\text{mol Trolox /g FW}$. Similarly, Flavonoids (Flav) and Glutathione (GSH) showed the most pronounced and significant increases under HA treatment (0.56 mg Quercetin /g FW and 0.11 $\mu\text{mol /mg protein/min}$, respectively), with maximum enhancement noted in the combined AMF+HA treatment (0.69 mg Quercetin /g FW and 0.13 $\mu\text{mol /mg protein/min}$, respectively), (Fig. 1).

For GSH specifically, a significant increase was observed under both HA and AMF+HA treatments, though this increase was less pronounced compared to other antioxidants. While there was no significant difference in GSH between AMF and HA individual treatments, the combined AMF+HA treatment resulted in a moderate yet significant increase, with a lesser effect observed under AMF or HA alone.

Regarding phenolic compounds, both AMF and HA treatments increased their content, with AMF showing the most significant enhancement. The combined AMF+HA treatment produced the highest phenolic content (12.84 mg GA /g FW), which clearly indicating a synergistic effect.

In summary, HA treatments generally enhanced TAC, Flavonoids, ASC, and GSH to a greater extent than AMF. Crucially, the combined AMF+HA treatment consistently yielded the highest and most significant levels of these measured antioxidants.

3.6. The effects of AMF, HA, and their combination on organic acid contents

Oxalic acid content displayed an increasing trend across all treatments, though these differences were not statistically significant compared to the control (Fig. 2). This indicates that while AMF, HA, and their combination influenced oxalic levels, they didn't drastically alter its accumulation in maize grains. In contrast, malic acid showed a more pronounced response. Both AMF and the combined AMF+HA treatments significantly increased malic content (17.21 and 16.56 $\mu\text{g /g FW}$, respectively) compared to the control. HA treatment also increased malic content, though less significantly. Interestingly, the minor decrease in malic acid observed under HA alone was less pronounced than in the control, suggesting HA might help mitigate malic degradation during maize grain development (Fig. 2).

For Succinic acid, all treatments significantly increased content relative to the control. The combined AMF+HA treatment produced the most significant increase (5.51 $\mu\text{g /g FW}$), followed by AMF alone, with HA showing the least increase. This demonstrates a clear synergistic effect from combined AMF+HA application, enhancing succinic levels more than either treatment individually.

For citric acid, Both AMF and the combined AMF+HA treatments significantly increased citric content compared to the control. The highest content was observed in AMF (4.45 $\mu\text{g /g FW}$), followed by the

Table 4

Elemental content in maize grains as influenced by AMF, HA, and their combined application.

Treatments	The content of element in grains ($\mu\text{g/g DW}$)							
	P	K	Ca	Mg	Na	Fe	Mn	Zn
Control	21.29 $\pm$ 2.71b	12.65 $\pm$ 2.80b	2.51 $\pm$ 0.32ab	1.77 $\pm$ 0.31a	0.76 $\pm$ 0.10c	0.27 $\pm$ 0.03b	0.15 $\pm$ 0.04b	0.61 $\pm$ 0.12b
AMF	16.01 $\pm$ 2.05c	15.71 $\pm$ 3.14ab	2.94 $\pm$ 0.39ab	1.90 $\pm$ 0.24a	1.21 $\pm$ 0.15ab	0.42 $\pm$ 0.05a	0.30 $\pm$ 0.04a	0.82 $\pm$ 0.29b
HA	10.52 $\pm$ 1.32d	13.93 $\pm$ 1.34b	2.46 $\pm$ 0.30b	1.79 $\pm$ 0.23a	1.02 $\pm$ 0.13b	0.30 $\pm$ 0.04b	0.20 $\pm$ 0.02b	0.78 $\pm$ 0.10b
AMF+HA	27.49 $\pm$ 3.62a	19.06 $\pm$ 2.50a	3.20 $\pm$ 0.41a	2.25 $\pm$ 0.22a	1.33 $\pm$ 0.16a	0.45 $\pm$ 0.06a	0.37 $\pm$ 0.05a	1.38 $\pm$ 0.17a

Data included are means $\pm$ SD; different letters within the same column indicate significant differences between means after Duncan's test ($P \leq 0.05$). AMF, Arbuscular mycorrhizal fungi; HA, Humic acid.

combined treatment (3.08 $\mu\text{g/g FW}$).

Regarding isobutyric, and fumaric acids, the results were less pronounced. No significant differences were observed across treatments for isobutyric acids, except for HA, which induced a significant increase (3.39 $\mu\text{g/g FW}$) compared to the control (2.54 $\mu\text{g/g FW}$). Fumaric acid showed a similar pattern, with a slight but significant increase observed under HA treatment (0.30 $\mu\text{g/g FW}$). These findings indicate that HA may have a more specific and significant effect on the accumulation of certain organic acids (isobutyric and fumaric acids) compared to AMF or the combined treatments (Fig. 2). Overall, the combined AMF+HA treatments generally resulted in the highest concentrations of oxalic and succinic acids, and AMF alone generally increase malic and citric acids, while HA alone was particularly effective in significantly increasing isobutyric, and fumaric acids.

3.7. Evaluation of AMF, HA, and their combined application impact by Principal Component (PCA)

All mean values of sugar metabolism and metabolite profiling related parameters of maize grains were analyzed by principal component analysis, PCA (Fig. 3). The first two principal components explained a cumulative 85.7 % of the total variance, with PC1 accounting for 58.7 % and PC2 for 27.1 %. This indicates that most of the variability among treatments is effectively captured in two dimensions.

4. Discussion

We explored how AMF and HA interact to influence maize grain biochemistry. By measuring primary metabolite accumulation, we aimed to demonstrate their role as sources for secondary metabolites, ultimately connecting this process to an enhanced biological value, notably the antioxidant capacity, of maize grains. To this end we performed a deep primary metabolic profiling (sugars, organic acids, amino acids and fatty acids) in maize grains. Then we link the changes in these primary metabolites to secondary antioxidant biosynthesis (phenols, flavonoids, tocopherols, ASC and GSH), to correlate these metabolic shifts with improved grain quality (total antioxidant capacity and mineral content), especially under low chemical fertilization. We hypothesize that AMF and HA synergy will optimize metabolite flow, thereby enhancing grain quality and associated biological activities. Our rigorous analysis of these bioactive compounds and minerals will elucidate the biochemical bases for improved maize performance and nutritional benefits.

4.1. AMF and HA synergistically enhance maize grain primary metabolite profiles

Our results indicate differential impacts, with AMF and HA contributing both distinct and synergistic effects, depending on the biochemical compound analyzed. Focusing initially on sugar metabolism, the research analyzed various sugar types in soil-grown maize grains, including glucose, fructose, sucrose, raffinose, cellulose, hemicellulose, and storage compounds (fructan, starch) (Table 1). The application of AMF alone significantly enhanced the accumulation of

sucrose (Begum et al., 2020) and lactose in maize grains compared to untreated samples. Similar increases were observed in glucose and fructose with independent HA treatment (Makhoulouf et al., 2022). Raffinose also showed the strongest increase with the combined treatment. While cellulose and hemicellulose were largely unaffected, and fructan accumulation notably increased under combined AMF+HA (Habib, 2021). This superior outcome with dual treatment highlights that AMF and HA synergistically optimize sugar metabolism in maize grains.

The increased levels of these essential sugars, especially glucose and fructose, are paramount as they directly provide the foundational carbon skeletons for synthesizing other critical primary metabolites (like amino acids and fatty acids) and serve as direct precursors for robust secondary metabolite production, thereby establishing a strong link to overall grain quality. In our previous study, the increased sugar availability, as discussed above, provided an enriched carbon source that strongly supported amino acid synthesis (Sheteiwy et al., 2023). This study revealed significant improvements in amino acid metabolism, with differential responses observed among various amino acids, suggesting specific regulatory mechanisms and metabolic pathways influenced by these signaling molecules. Our analysis differentiated between essential and non-essential amino acids due to their varying nutritional importance for humans and their diverse roles in protein synthesis and stress responses within the plant, thereby directly affecting the protein quality and overall biological value of the maize grain. The superior efficacy of individual AMF treatment in enhancing Lysine and Alanine concentrations is particularly noteworthy. This finding aligns with previous research by Puccio et al. (2023), who reported AMF-induced upregulation of amino acid biosynthesis in wheat. The enhanced accumulation of Lysine, an essential amino acid often limited in cereals, significantly improves grain nutritional quality (Table 3), consistent with AMF effects reported in tomato (Carillo et al., 2020) and onion (Metwally et al., 2021). HA remarkably affected Histidine, Methionine, Threonine, Valine, and Asparagine, suggesting targeted metabolic activation, aligning with Galambos et al. (2020). Conversely, Isoleucine decreased across all treatments except for AMF. Interestingly, combined AMF+HA showed no additive/synergistic effects for several amino acids, implying potential metabolic interference (Table 3). Individual treatment-specific amino acid increases are likely due to mechanisms like AMF's modulation of nitrogen metabolism and biosynthesis via key enzymes (Wu et al., 2024), suggesting independent or competitive operations. These findings are important for understanding hormone-mediated amino acid regulation in cereal crops.

Moving to fatty acid metabolism, our findings reveal complex interactions between these signaling molecules and fatty acid metabolism. We specifically examined both saturated and unsaturated fatty acids due to their distinct roles in energy storage, membrane structure, and their critical impact on the nutritional profile and functional properties of maize grains, which directly influence the bioavailability and overall biological activity of the grain. The observed increase in saturated fatty acids, particularly Palmitic and Stearic acids, under both individual and combined treatments aligns with previous research in soybeans (Marro et al., 2020; Igiehon et al., 2021) and maize (Alsherif et al., 2022). This increase may be attributed to AMF's role in upregulating fatty acid

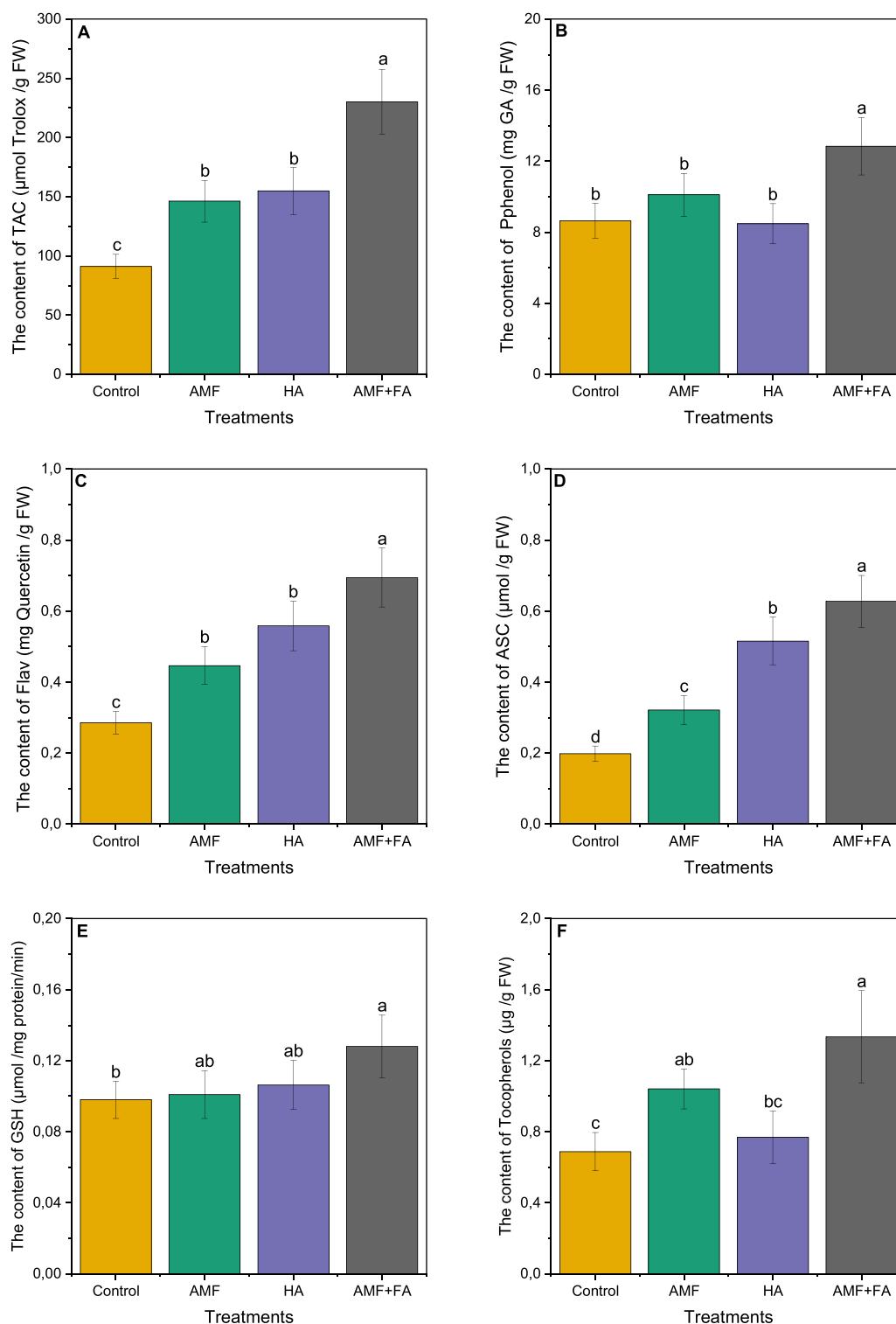


Fig. 1. Antioxidant content in maize grains as influenced by AMF, HA, and their combined application. (A) Total antioxidant capacity (TAC); (B) Total phenols; (C) Flavonoids; (D) Ascorbate (ASC); (E) Glutathione (GSH); (F) Tocopherols. Values represent means $\pm$ SD. Different letters indicate significant differences among treatments according to Duncan's multiple range test ($P \leq 0.05$).

synthase genes, as demonstrated by [Zhang et al. \(2024\)](#) in their studies. Regarding unsaturated fatty acids, the significant increase in palmitoleic, linoleic, and eicosenoic acids under combined treatment is particularly noteworthy ([Table 2](#)). These findings are consistent with [Ezzati Lotfabadi et al., \(2022\)](#), who observed similar increases in unsaturated fatty acids following AMF treatment in soybean. The enhancement of these essential fatty acids could have important implications for grain

nutritional quality, as suggested by recent studies ([Nasiroleslami et al., 2021](#)). The enhanced accumulation of both saturated and unsaturated fatty acids under combined AMF+HA treatment could be attributed to several mechanisms. First, AMF has been shown to enhance the expression of fatty acid biosynthesis genes ([Zhang et al., 2024](#)). Second, HA is known to modulate membrane lipid composition and fatty acid desaturase activity ([Liu et al., 2023](#)). The synergistic effect observed in

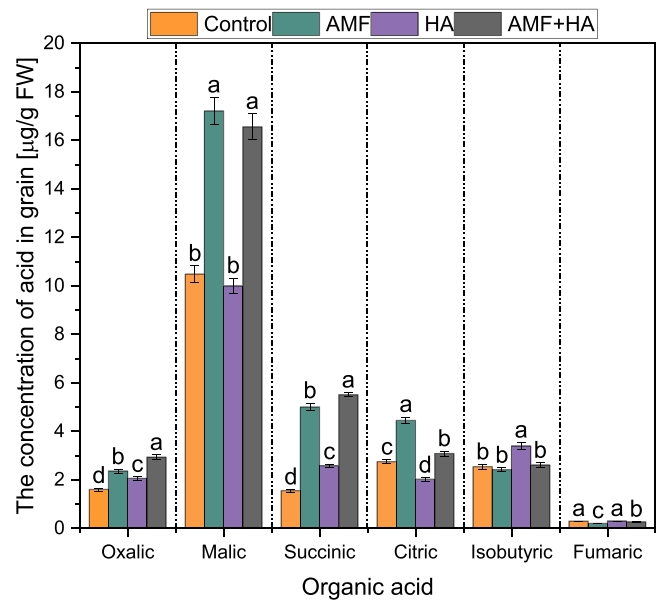


Fig. 2. Organic acid content in maize grains as influenced by AMF, HA, and their combined application. Values represent means $\pm$ SD. Different letters indicate significant differences among treatments according to Duncan's multiple range test ($P \leq 0.05$).

our study suggests these mechanisms may be complementary, leading to enhanced fatty acid accumulation when both compounds are applied together.

Finally, our findings on organic acid accumulation further

underscore the broad impact of AMF and HA on primary metabolism, influencing key intermediates crucial for energy cycles and precursor supply. Organic acid accumulation was distinctly impacted by AMF and HA (Fig. 2). Oxalic acid changes were minimal. Malic acid was significantly enhanced by AMF alone and AMF+HA (Amani Machiani et al., 2022), with HA having a lesser effect. Succinic acid increased across all treatments, with AMF+HA showing the strongest synergistic boost (El-Kinany et al., 2020). AMF alone significantly increased citric acid, while HA alone significantly increased isobutyric, and fumaric acids, consistent with its role in stress metabolism (Chen et al., 2022). Overall, AMF predominantly enhanced organic acids.

4.2. AMF and HA synergistically enhance maize grain secondary metabolite profiles

The significant enhancements observed in primary metabolites, sugars, fatty acids, and amino acids are not merely increases in content but represent fundamental improvements that serve as crucial building blocks and precursors for the synthesis of vital secondary metabolites and contribute directly to the enhanced biological value of maize grains. This study showed increases phenolic in secondary compounds including flavonoids, ascorbic acid, glutathione, and tocopherols under AMF and HA (alone or combined) (Fig. 1). These increases in antioxidant compounds are directly linked to the enhanced primary metabolite pool (e.g., sugars, organic acids) discussed previously, which serve as crucial precursors. Flavonoid and ascorbic acid levels were also enhanced, especially with AMF+HA, supporting the role of HA in activating flavonoid and ascorbate pathways (Zaremanesh et al., 2021; Liu et al., 2022; Abu-Ria et al., 2024), pathways that largely rely on carbon skeletons derived from central primary metabolism. For Glutathione (GSH), HA alone led to notable increases, although the combined

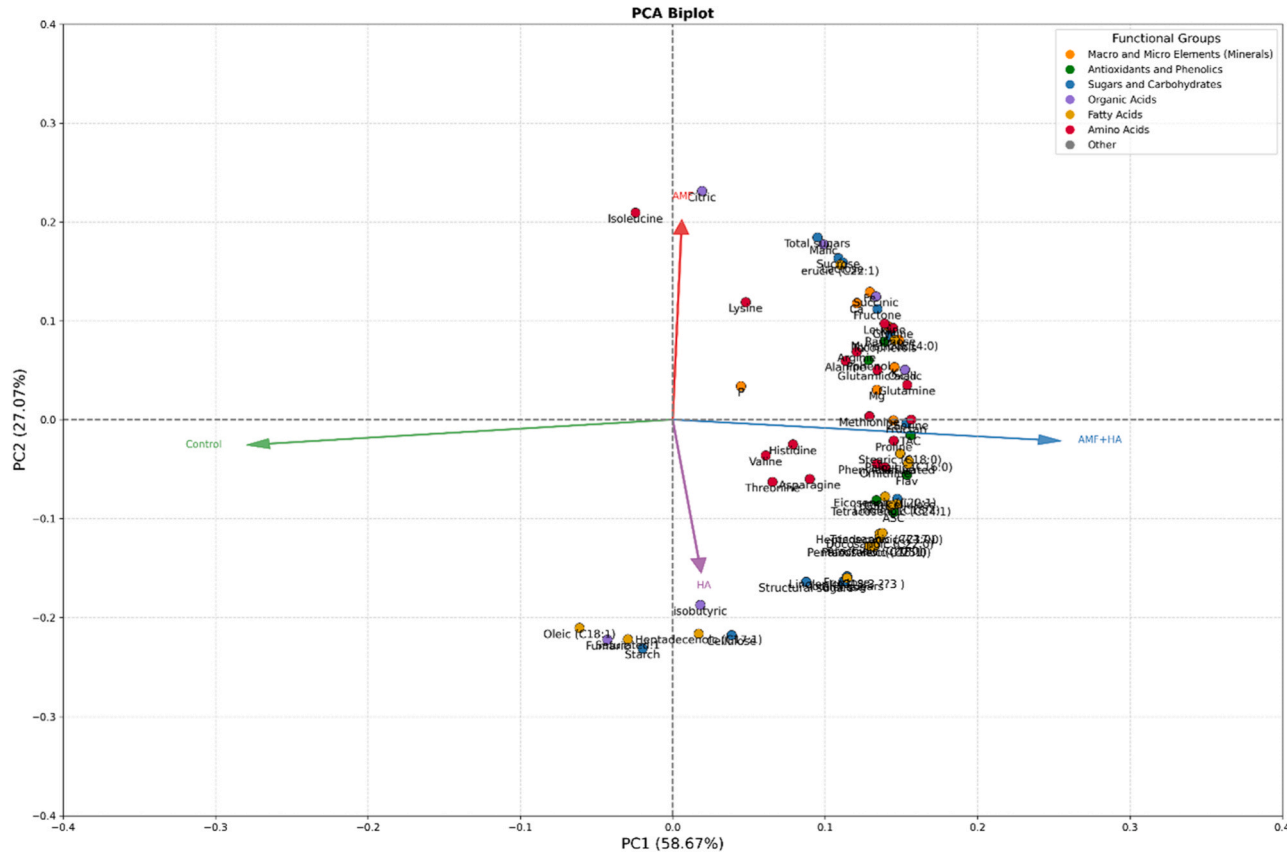


Fig. 3. Principal component analysis (PCA) showing the changes in sugar metabolism and metabolite profiling related parameters of maize grains as influenced by AMF, HA, and their combined application.

treatment produced a more moderate effect compared to TAC and ASC (Fig. 1). Phenolic compounds increased most significantly under AMF, with the AMF+HA combination further boosting levels, emphasizing AMF's role in phenolic synthesis (Ran et al., 2022), a process that utilizes phenylalanine, an amino acid derived from the shikimate pathway which initiates from primary carbon metabolism. Similarly, Tocopherol content peaked under AMF+HA, supporting research by Ozturk et al., (2023), which links AMF to tocopherol biosynthesis and cellular protection. In conclusion, HA and AMF each increased TAC, Flavonoids, and ASC, while AMF was especially effective for Phenolic compounds and Tocopherols.

4.3. Enhanced maize grain quality: Linking primary metabolites with bioactive secondary, antioxidant capacity and essential minerals

The enhanced accumulation of primary metabolites induced the accumulation of secondary, bioactive compounds. In addition to these bioactive compounds, such as essential amino acids and unsaturated fatty acids which possess significant biological value, we observed increased antioxidant metabolic capacity (TAC) and improved essential mineral content. This multifaceted improvement highlights how AMF and HA contribute to the overall nutritional and biological value of maize grains. TAC showed the highest increase under the combined AMF+HA treatment, suggesting a synergistic effect on overall antioxidant capacity, as observed in recent studies (Aalipour et al., 2020; Sun and Shahrajabian, 2023).

Furthermore, these treatments notably influenced the essential mineral composition of maize grains, directly impacting their nutritive and biological value. A significant reduction in P content under AMF treatment, amplified by HA, suggested unique metabolic interactions (Table 4). However, the combined AMF+HA treatment later increased P content, aligning with findings of counterbalancing effects on nutrient accumulation (Cuzzolino et al., 2021). This modulated phosphorus is crucial for energy transfer and nucleic acid synthesis, directly impacting grain vitality (Malhotra et al., 2018). In contrast, K content consistently increased across all treatments, peaking with AMF+HA, supporting enhanced K uptake and transport by these amendments (Aalipour et al., 2020). Potassium is vital for enzyme activation and osmotic regulation, contributing significantly to overall grain health and nutritional profile (Johnson et al., 2022). For Ca, Mg, Na, Fe, Mn, and Zn, there was a differential response of these elements to the treatment groups compared to the control (Table 4), indicating the influence of AMF+HA on their accumulation (Table 4).

5. Conclusion

This report highlights that individual and combined AMF and HA applications significantly promote specific sugar metabolism and overall biochemical profiling in maize grains. Crucially, this primary metabolite accumulation enhanced secondary metabolite production, thereby increasing the overall biological value of the grain. The combined treatment synergistically enhanced vital sugars (sucrose, glucose, fructose) and HA specifically increased certain organic acids (isobutyric, fumaric), markedly improving the overall biochemical profile and suggesting potential for boosting crop yield and quality in sustainable agriculture. However, study limitations include focus on a specific maize variety and environmental conditions, alongside a lack of deeper molecular mechanistic insight. Future research should assess the long-term impacts on soil fertility, crop sustainability, and applicability to other crops.

CRediT authorship contribution statement

Mohamed S. Sheteiwy: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Visualization, Investigation, Supervision, Software, Validation, Writing – review & editing. **Ahmed**

M. El-Sawah: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Visualization, Investigation, Supervision, Software, Validation, Writing – review & editing. **Mohammed Alyafei:** Data curation, Writing – original draft, Writing – review & editing. **Saad Sulieman:** Data curation, Writing – original draft. **Basharat Ali:** Visualization, Investigation. **Izabela Joško:** Visualization, Investigation, Writing – review & editing. **Mikołaj Feculak:** Visualization, Investigation. **Ali El-Keblawy:** Visualization, Investigation, Supervision. **Zaid Ulhassan:** Visualization, Investigation. **Yousef Alhaj Hamoud:** Software, Validation. **Hiba Shaghaleh:** Software, Validation. **Mohammad K. Okla:** Software, Validation, Writing – review & editing. **Ibrahim A. Alaraidh:** Software, Validation, Writing – review & editing. **Ghada G. Abdel-Fattah:** Conceptualization, Methodology, Software, Data curation, Writing – original draft, Visualization, Investigation, Supervision, Software, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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